AUDITLENS: QR-BASED MACHINE SAFETY INSPECTION AND LOCKOUT SYSTEM

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

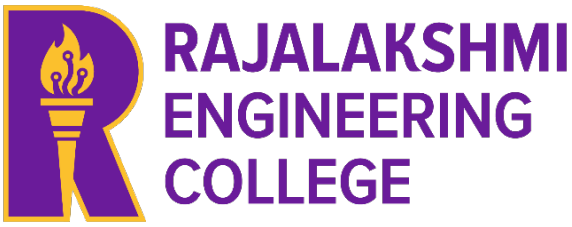
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND BUSINESS SYSTEMS



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(An Autonomous Institution, Affiliated to Anna University, Chennai)

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled “AUDITLENS: QR-BASED MACHINE SAFETY INSPECTION AND LOCKOUT SYSTEM” is the bonafide work of SIVARATHINAM S (2116241401102), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, SIVARATHINAM S (2116241401102), hereby declare that the project report titled “AUDITLENS: QR-BASED MACHINE SAFETY INSPECTION AND LOCKOUT SYSTEM”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts and workflows — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 03 November 2026

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ABSTRACT

Factories must keep guards fitted on dangerous machine parts and inspect machines on a fixed schedule. This is usually tracked on paper and Excel, so inspection dates are missed, missing guards go unnoticed and unsafe machines keep running. During an inspector's visit, records are hard to find, and nobody can prove that machines were inspected and safe to operate.

This project develops AuditLens, a QR-based machine safety inspection and lockout system on Zoho Creator, a low-code platform chosen because it provides forms, a managed database, reports, pages and workflow automation without building and hosting a full software stack. The application has three forms (Technician, Machine and Inspection_Log), a set of list, board and chart reports including an Inspector report, and one Safety Dashboard page with four cards and four charts. Every machine receives a QR code that links to its live record. Deluge form workflows calculate inspection due dates, lock out a machine whose guard is missing or damaged, and update the machine after each inspection. An unsafe result locks the machine out and e-mails the responsible technician. A daily scheduled workflow locks out overdue machines and marks machines due within seven days as Due Soon. The application was published from Development to Stage and Production as version 1.0.

The application was tested with a set of 13 machines. Errors found during development, such as mismatched field link names, a wrong workflow event and import values that did not match drop-down spelling, were traced and fixed. AuditLens replaces paper and Excel tracking with automatic due dates, lockout, alerts and an audit-ready inspection history.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; QR Code; Machine Safety; Inspection Management; Workflow Automation*

PROJECT AT A GLANCE

Attribute	Details
Project Title	AuditLens: QR-Based Machine Safety Inspection and Lockout System
Application Domain	Industrial Safety and Compliance / Machine Inspection Management
Student Name & Register No.	SIVARATHINAM S – 2116241401102
Zoho Creator Edition	Zoho Creator (trial edition)
Application Name (Link Name)	AuditLens (audit-lens)
Published / Live URL	https://creatorapp.zoho.in/sivarathinam.s.2024.csbs1648/audit-lens/
Demo Credentials (per role)	Admin login: sivarathinam.s.2024.csbs@rajalakshmi.edu.in (password provided separately). Technicians are records in the Technician form; separate role-wise logins are not used in this version.
Number of Forms / Subforms	3 forms (Technician, Machine, Inspection_Log) / 0 subforms
Number of Reports (by type)	List: 4 (All Machines, Locked Out Machines, Inspection_Log Form Report, Inspector report) Kanban: 1 (Machine Status Board) Chart: 2 reports (Machines by Section, Inspections by Result) and 4 charts on the dashboard page Calendar: 0
Number of Pages / Dashboards	1 (Safety Dashboard)
Number of Workflows	Form workflows on Machine and Inspection_Log Schedule: 1 (Daily Safety Check) Approval: 0 Blueprint: 0 Batch: 0
Deluge Functions / Custom Actions	5 Deluge scripts (DS-01 to DS-05) in form workflows and 1 schedule; 0 standalone functions
Integrations	Zoho Mail (sendmail lockout alerts); no external API
Roles & Profiles Defined	Zoho Creator login; separate permission profiles are proposed for a future version (Administrator, Technician, Safety Manager, Auditor)
Environments Used	Development, Stage and Production (trial account), version 1.0
Date of Production Deployment	02 November 2026 (version 1.0 in Stage and Production)

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSV	Comma-Separated Values
ER	Entity–Relationship
FR	Functional Requirement
IoT	Internet of Things
LCAP	Low-Code Application Platform
OSHA	Occupational Safety and Health Administration
PDF	Portable Document Format
QR	Quick Response (code)
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SMS	Short Message Service
TC	Test Case
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Factories use machines with moving parts, pinch points, hot surfaces and stored energy. Guards must stay fitted on these dangerous parts, and each machine must be inspected on a fixed schedule so that wear, damage and missing safety devices are found before someone is hurt. In India, the Factories Act, 1948 requires dangerous parts of machinery to be securely fenced [3], and international practice such as OSHA 29 CFR 1910.147 requires hazardous energy to be controlled before servicing a machine [4].

In most small and medium plants this work is tracked on paper checklists and Excel sheets. Dates are written by hand, nobody is reminded automatically, and a machine with a missing guard can keep running until the next person notices. This project builds a digital system for this process on Zoho Creator. Every machine gets a QR code; the application calculates inspection due dates, locks out unsafe machines, e-mails the responsible technician and keeps the inspection history as audit proof.

1.2 Problem Statement

Inspection and guard checks are tracked manually, so the plant cannot show at any time which machines are inspected, which are overdue and which are unsafe to run.

- Inspection dates are missed because nobody is reminded automatically.
- Missing or damaged guards go unnoticed until someone is hurt.
- Unsafe machines keep running because nobody is forced to stop them.
- During an inspector’s visit, records are hard to find.

In one line: No one is automatically reminded, held accountable, or able to prove that machines were inspected and safe to operate.

1.3 Objectives

- To register every machine once, with a unique QR tag.
- To calculate inspection due dates automatically.
- To lock out machines that are unsafe, have a missing guard, or are overdue.
- To notify the responsible technician by e-mail.
- To store inspection history as audit evidence.
- To show the overall safety status on a dashboard.

1.4 Scope of the Project

In scope: machine registry, QR codes, inspection log, automatic due dates, automatic lockout, e-mail alerts, reports and a dashboard.

Out of scope (future work): WhatsApp/SMS alerts, predictive maintenance, IoT sensors and ERP integration. The lockout in this application is a status in the software that is shown to every user; it does not cut power to the physical machine.

1.5 Justification for a Low-Code Approach

AuditLens is a typical departmental business application: three forms, clear rules, e-mail notifications, reports and a dashboard, used by non-technical shop-floor staff. With traditional code it would need a database, a back-end, a web front-end, an e-mail service, QR generation and hosting. Zoho Creator provides these as managed services; forms, reports and pages are built visually and Deluge, Zoho's scripting language, is used only where custom logic is needed [1], [2]. Low-code platforms are widely recognised for this class of data-centric application [5], [6].

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The current process is manual. A technician inspects a machine and records the result on a paper checklist. The date is later copied into an Excel sheet, and a supervisor looks at the sheet to decide which machines are due. Guard condition is noted only when someone remembers to check it. When an inspector visits, the paper records and the sheet are searched by hand. Figure 2.1 shows this as-is process.

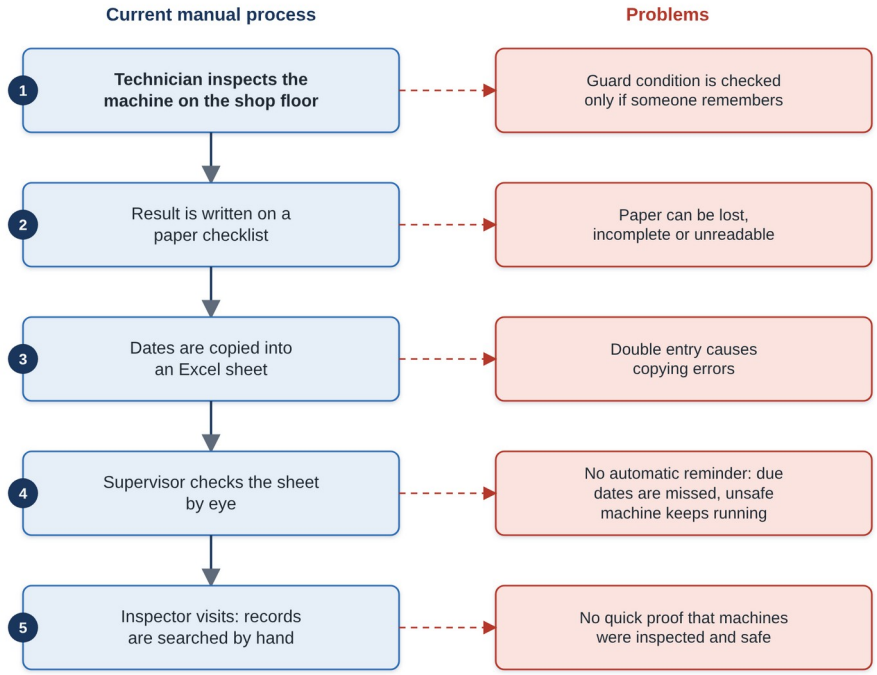


Figure 2.1: As-Is Process Flow of the Existing Inspection System

2.2 Limitations of the Existing System

- Data is written twice (paper, then Excel), which wastes time and introduces copying errors.
- There is no automatic reminder, so inspection dates are missed.
- A missing or damaged guard has no immediate consequence; the machine keeps running.
- No one is alerted when a machine fails an inspection.
- Records are scattered, so proving compliance to an inspector is slow.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development effort	Database, back-end, front-end and deployment built separately	Forms, reports, page and workflows built in one platform
Skills required	SQL, a back-end language, HTML/CSS/JavaScript, DevOps	Zoho Creator builders and basic Deluge scripting
Hosting and infrastructure	Server or cloud hosting, database administration, backups	Hosted and backed up by Zoho
QR and e-mail	Separate QR library and mail service to integrate	QR/Barcode field and sendmail task built in
Maintenance and change	Code changes, testing and redeployment	Fields, reports and workflows edited visually
Customisation limits	Unlimited	Limited to platform features; complex UI needs widgets or pro-code
Cost	Developer time plus hosting	Per-user subscription (trial edition used for this project)

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Sahay, Indamutsa, Di Ruscio and Pierantonio, IEEE SEAA, 2020	Supporting the understanding and comparison of low-code development platforms	Compares low-code platforms and their common features: visual modelling, data management, workflow and integration.	Guided the choice of Zoho Creator features for forms, workflows and e-mail.
2	Bock and Frank, Business & Information Systems Engineering, 2021	Low-Code Platform	Defines low-code platforms and their typical use for business applications by citizen developers.	Supports the justification for a low-code approach.
3	Di Ruscio, Kolovos, de Lara, Pierantonio, Tisi and Wimmer, Software and Systems Modeling, 2022	Low-code development and model-driven engineering: Two sides of the same coin?	Relates low-code platforms to model-driven engineering and discusses their limitations.	Used in the discussion of limitations and the pro-code spectrum.
4	Government of India, 1948	The Factories Act, 1948 (fencing of machinery)	Requires dangerous parts of machinery to be securely fenced.	Basis for the guard-condition check and the lockout rule.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
5	OSHA, U.S. Department of Labor	29 CFR 1910.147, Control of Hazardous Energy (Lockout/Tagout)	Requires hazardous energy to be isolated and machines locked out before servicing.	Concept behind locking out unsafe machines.
6	Zoho Corporation, online documentation	Zoho Creator Help and Deluge Reference	Documents forms, reports, pages, workflows, QR fields and sendmail.	Primary reference for implementing every component.

2.5 Emerging Trends in Low-Code Development

Several current trends shaped the design. Citizen development is reflected in keeping the application simple enough for a safety officer to maintain. Hyper-automation is reflected in chaining the due-date calculation, lockout and e-mail alert on a single inspection submission. Composable applications are reflected in reusing Zoho Mail for alerts instead of building a mail service. AI-assisted building and Zia features were not used in this project; they are listed as future enhancements.

2.6 Proposed System

The proposed system replaces the paper checklist and Excel sheet with one Zoho Creator application. Each machine is registered once and receives a QR code that links to its live record. Deluge workflows calculate due dates, lock out machines with a missing or damaged guard, and update the machine after every inspection. An unsafe result locks the machine out and e-mails the assigned technician. A dashboard and reports read the same data, and the Inspection_Log keeps the history as audit proof. Each limitation listed in Section 2.2 is therefore addressed directly.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Stakeholder	Responsibility	Use of AuditLens
Administrator	Maintains the machine registry and technicians	Registers machines, generates QR codes, reviews the dashboard
Technician	Inspects machines on schedule	Scans the QR code, submits inspections, receives lockout e-mails
Safety Manager / Inspector	Checks that machines are safe and inspected	Views the dashboard and inspection history as audit evidence

In this version these stakeholders use the application through their Zoho Creator logins. Separate permission profiles for each role are proposed in Section 4.6 and Chapter 9.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Status
FR-01	Register a machine with a unique Serial_Number and auto-numbered Machine_ID	Implemented
FR-02	Generate a QR code per machine that links to its live record	Implemented
FR-03	Convert Calibration_Frequency to Frequency_Days (Monthly 30, Quarterly 90, Half-Yearly 180, Yearly 365)	Implemented
FR-04	Calculate Next_Due_Date = last inspection date + Frequency_Days	Implemented
FR-05	Lock out a machine whose guard is Missing or Damaged	Implemented
FR-06	Record an inspection with result, guard check, e-stop test, certificate number, photo and remarks	Implemented
FR-07	Update the machine’s dates after each inspection	Implemented
FR-08	Lock out the machine when the result is Unsafe or the guard is not intact	Implemented
FR-09	E-mail the assigned technician on lockout	Implemented
FR-10	Return the machine to Active after a safe inspection with the guard intact	Implemented
FR-11	Open the machine’s live record when the QR code is scanned	Implemented
FR-12	Show cards and charts of overall safety status on a dashboard	Implemented
FR-13	Provide an All Machines report with status, risk, guard, QR and technician	Implemented

FR ID	Requirement	Status
FR-14	Daily schedule: lock out overdue machines; mark machines due within 7 days as Due Soon	Implemented
FR-15	Inspector report on Inspection_Log with filters by inspection date, machine and result	Implemented

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

Category	Requirement
Usability	A technician should reach a machine's record with one QR scan and submit an inspection on one form.
Reliability	Status and dates must always be calculated by the workflows, never typed by hand.
Data integrity	Serial_Number must be unique; mandatory fields and fixed drop-down values prevent bad data.
Maintainability	Logic is kept in a few short Deluge workflows that a non-expert can edit.
Availability	Hosted by Zoho Creator; accessible from a browser on a phone or desktop.
Security	Scanning the QR code requires a login to the application.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Item	Requirement
Platform	Zoho Creator (trial edition used)
Scripting	Deluge
Client device	Desktop or laptop with a modern web browser; smartphone with a camera for QR scanning
Network	Internet connection
Labels	Printer for QR labels attached to machines
Mail	Zoho Mail sender (trial sender address used)

3.5 Use Case Model

Figure 3.1 shows the actors and their main use cases. Table 3.5 describes the key use case in detail.

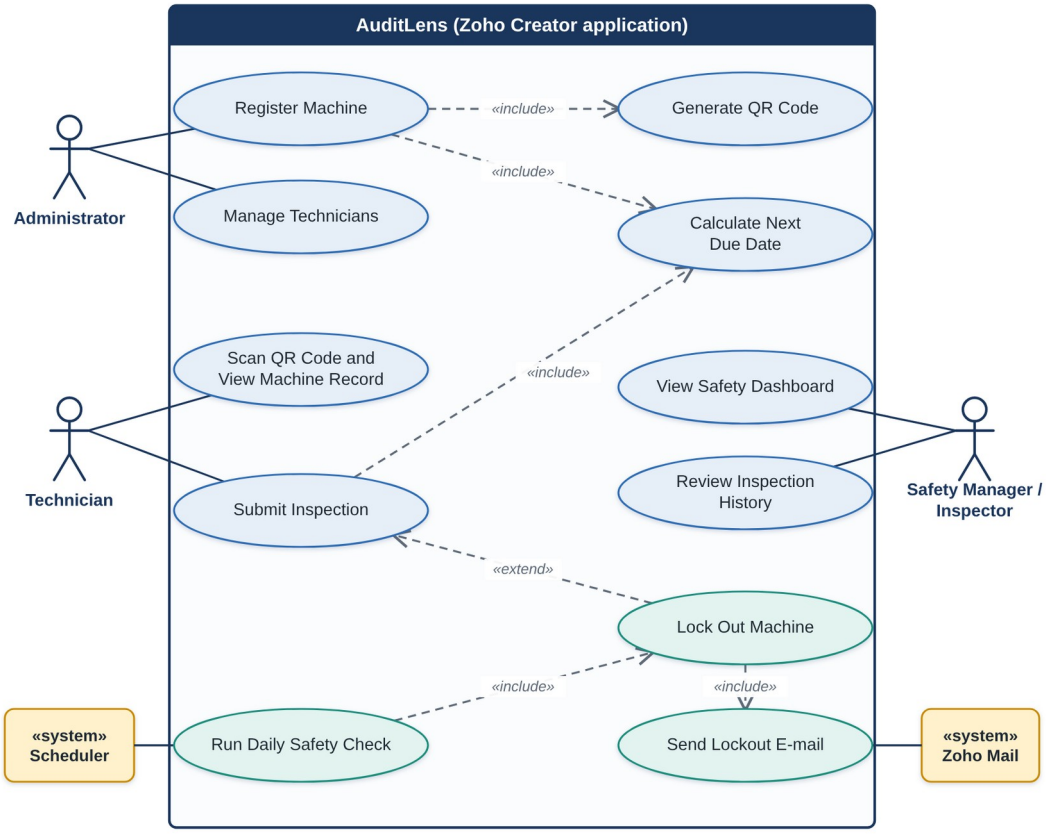


Figure 3.1: Use Case Diagram of AuditLens

Table 3.5: Use Case Description – Submit an Inspection and Lock Out an Unsafe Machine

Item	Description
Use Case ID	UC-01
Use Case Name	Submit an inspection and lock out an unsafe machine
Actor(s)	Technician (primary); Administrator and Zoho Mail (secondary)
Pre-condition	The machine is registered and has a QR code; the technician is logged in.
Main Flow	1. The technician scans the QR code and the machine record opens. 2. The technician opens the Inspection_Log form and selects the machine. 3. The technician enters the date, result, guard check, e-stop test, certificate number, photo and remarks. 4. On validation the next due date is calculated. 5. On success the machine's dates are updated. 6. If the result is Unsafe or the guard is not intact, the machine is locked out and an e-mail is sent to the assigned technician.
Alternate Flow	6a. If the result is safe and the guard is intact, the machine status is set to Active and the guard condition to Intact.
Post-condition	The inspection is stored as history; the machine shows the new status, dates and guard condition on the report and dashboard.
Related FR	FR-06 to FR-10

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

AuditLens runs entirely on Zoho Creator. Users reach it through a browser or by scanning a QR code. Three forms store the data; form workflows written in Deluge calculate dates and statuses and send the lockout e-mail through Zoho Mail; reports and one dashboard page read the same data. A daily scheduled workflow handles overdue machines (Section 5.6). Figure 4.1 shows the architecture.

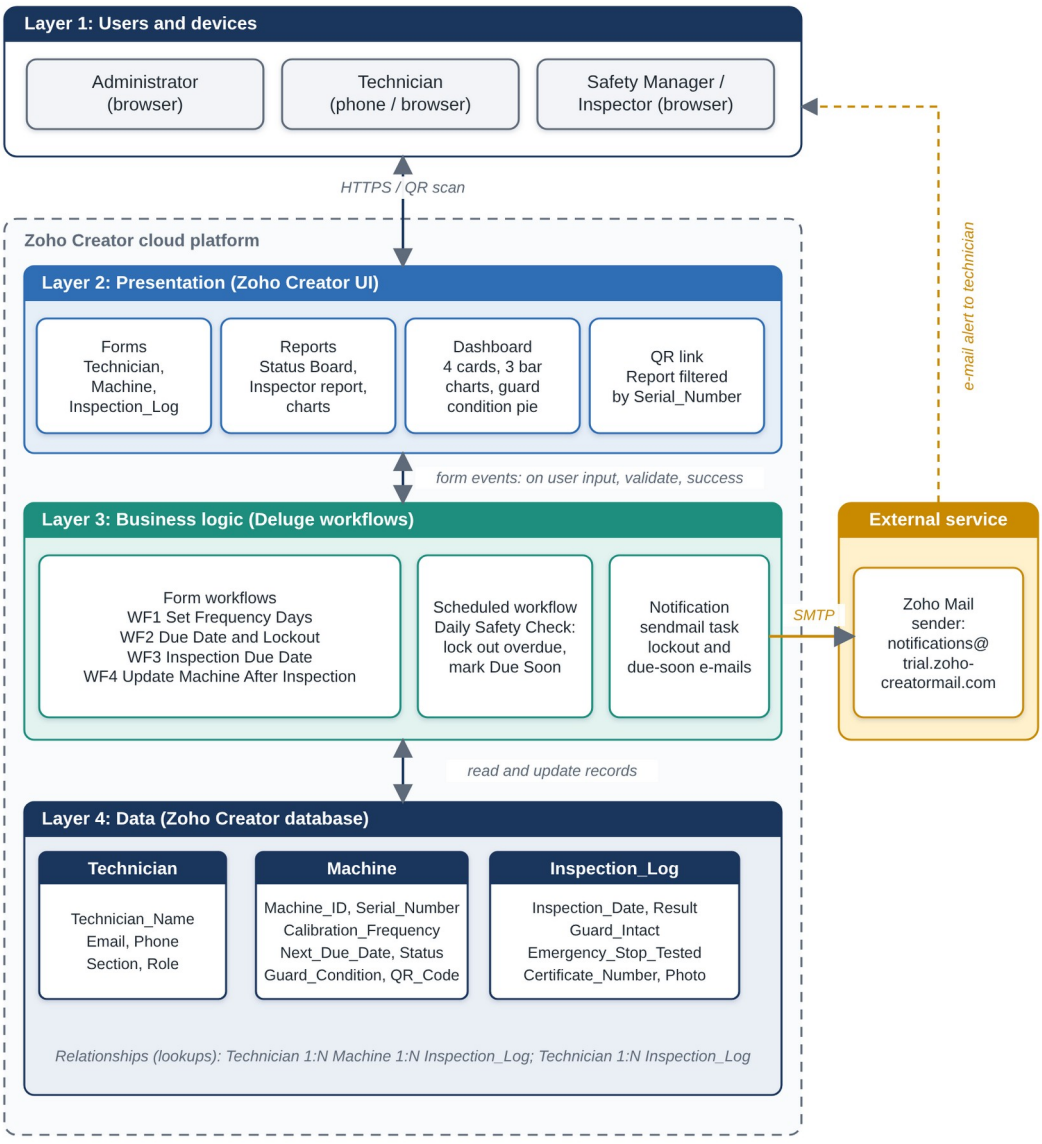


Figure 4.1: System Architecture of AuditLens

4.2 Data Model

The data model has three entities. A Technician is responsible for many machines; a Machine has many inspections; each inspection is performed by a technician. Relationships are implemented with lookup fields (Machine.Technician, Inspection_Log.Machine and Inspection_Log.Performed_By). Figure 4.2 shows the entity–relationship diagram and Table 4.1 lists the relationships.

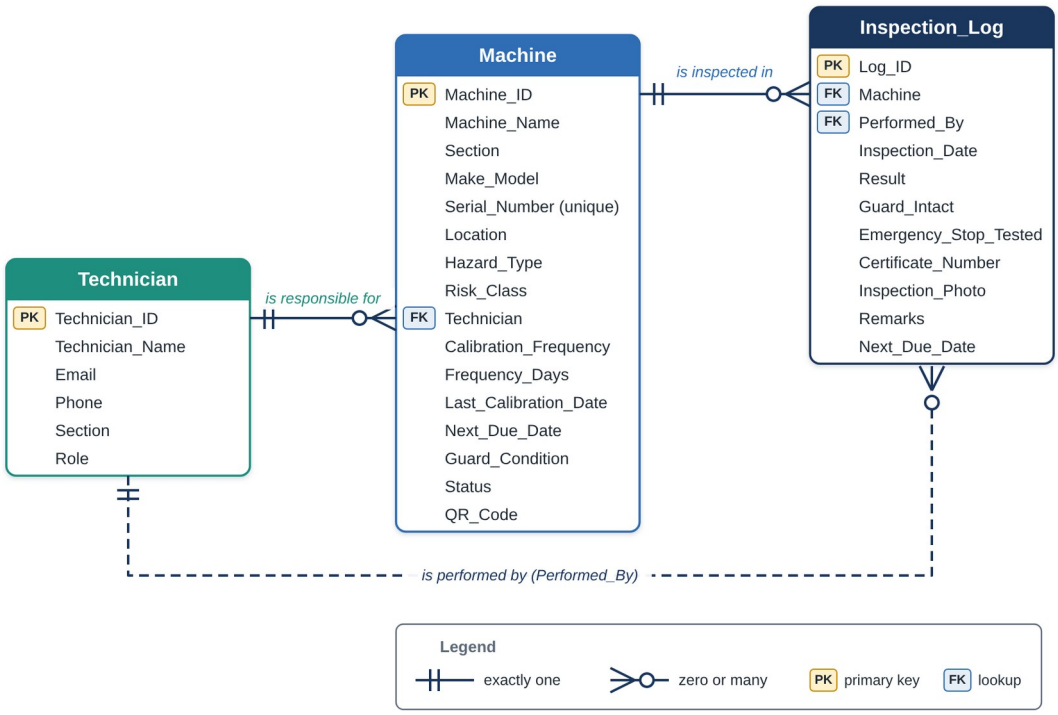


Figure 4.2: Entity–Relationship Diagram of the Application

Table 4.1: Entities and Relationships

Entity	Related Entity	Type	Implemented By
Technician	Machine	1 : N	Lookup field Technician on Machine
Machine	Inspection_Log	1 : N	Lookup field Machine on Inspection_Log
Technician	Inspection_Log	1 : N	Lookup field Performed_By on Inspection_Log

4.3 Form Design

Table 4.2 lists the fields of the three forms. Machine and Inspection_Log carry the calculated and workflow-controlled fields.

Table 4.2: Field Specification – AuditLens Forms

Form	Field	Type	Notes
Technician	Technician_Name	Single line	Mandatory
Technician	Email	Email	Used for alerts

Form	Field	Type	Notes
Technician	Phone	Phone	
Technician	Section	Single line	Plant section
Technician	Role	Radio	
Machine	Machine_ID	Auto number	Start index 1001; generated on save
Machine	Machine_Name	Single line	
Machine	Section	Single line	Link name Section1 in this form
Machine	Make_Model	Single line	
Machine	Serial_Number	Single line	Unique; used in the QR link
Machine	Location	Single line	
Machine	Hazard_Type	Single line	Used in a dashboard chart
Machine	Risk_Class	Radio	Low, Medium, Critical
Machine	Statutory_Inspection_Required	Radio	Yes / No
Machine	Technician	Lookup	Links to Technician form
Machine	Calibration_Frequency	Drop down	Monthly, Quarterly, Half-Yearly, Yearly
Machine	Frequency_Days	Number	Label: Inspection frequency; set by workflow
Machine	Last_Calibration_Date	Date	Label: Last Inspection Date
Machine	Next_Due_Date	Date	Set by workflow
Machine	Guard_Condition	Radio	Intact, Damaged, Missing
Machine	Status	Drop down	Active, Due Soon, Locked Out, Under Maintenance
Machine	QR_Code	QR / Barcode	Encodes link by Serial_Number
Inspection_Log	Machine	Lookup	Links to Machine form
Inspection_Log	Inspection_Date	Date	
Inspection_Log	Performed_By	Lookup	Links to Technician form
Inspection_Log	Result	Radio	Safe, Unsafe, Repaired and safe
Inspection_Log	Guard_Intact	Decision box	Unticked means lockout
Inspection_Log	Emergency_Stop_Tested	Decision box	
Inspection_Log	Certificate_Number	Single line	
Inspection_Log	Inspection_Photo	File upload	Evidence photo
Inspection_Log	Remarks	Multi line	
Inspection_Log	Next_Due_Date	Date	Set by workflow

4.4 Process and Workflow Design

Figure 4.3 shows where each workflow runs. When a machine is saved, the Machine workflows set Frequency_Days and Next_Due_Date and lock out the machine if the guard is Missing or Damaged. When an inspection is submitted, the Inspection_Log workflows calculate the next due date and then update the machine; an unsafe result locks the machine out and sends an e-mail.

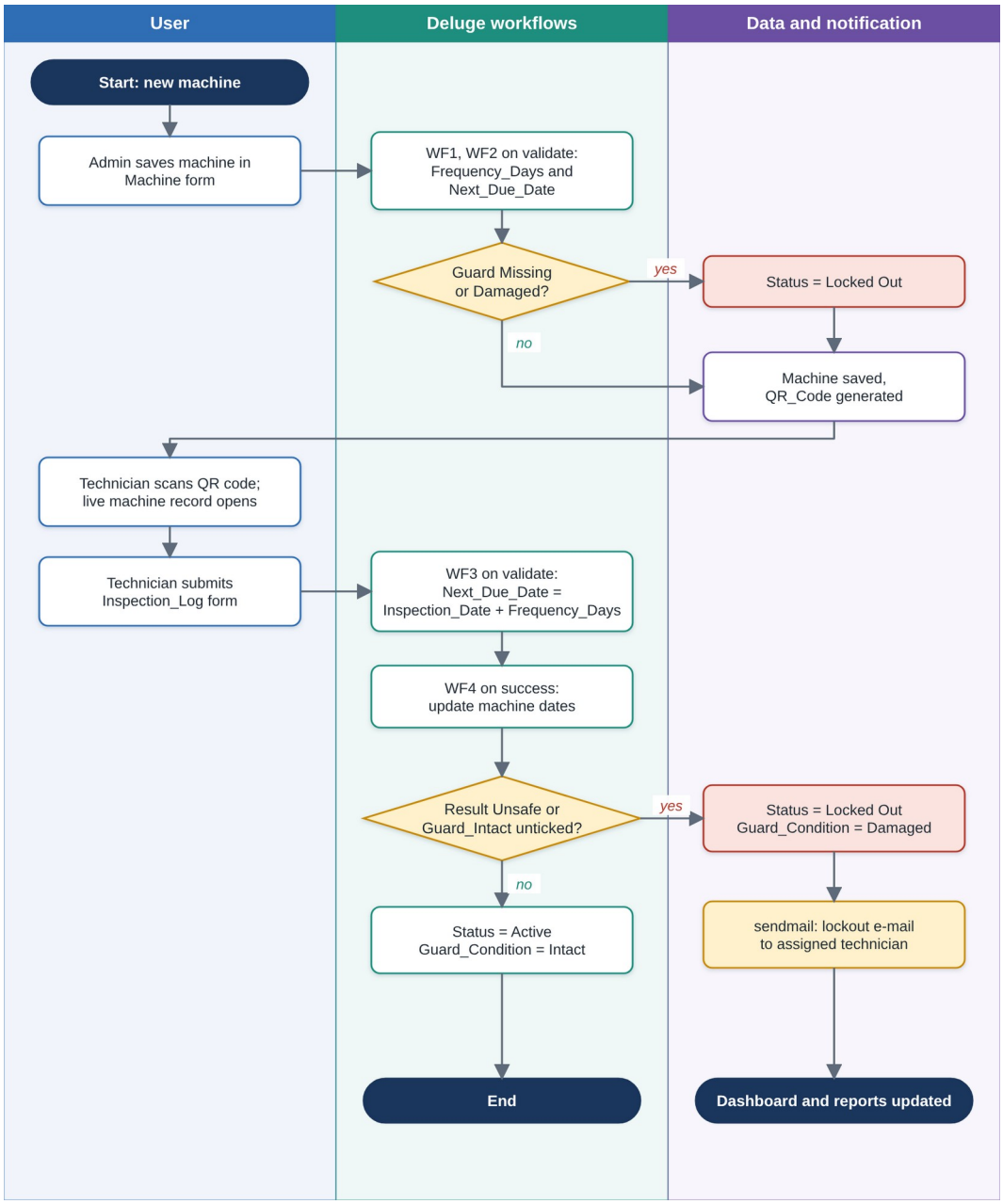


Figure 4.3: Workflow for Registering a Machine and Submitting an Inspection

4.5 User Interface and Navigation Design

The application has three forms, several reports and one dashboard page. The QR code links directly to the All Machines report filtered by Serial_Number, so a scan opens one machine's live record. Figure 4.4 shows the navigation structure.

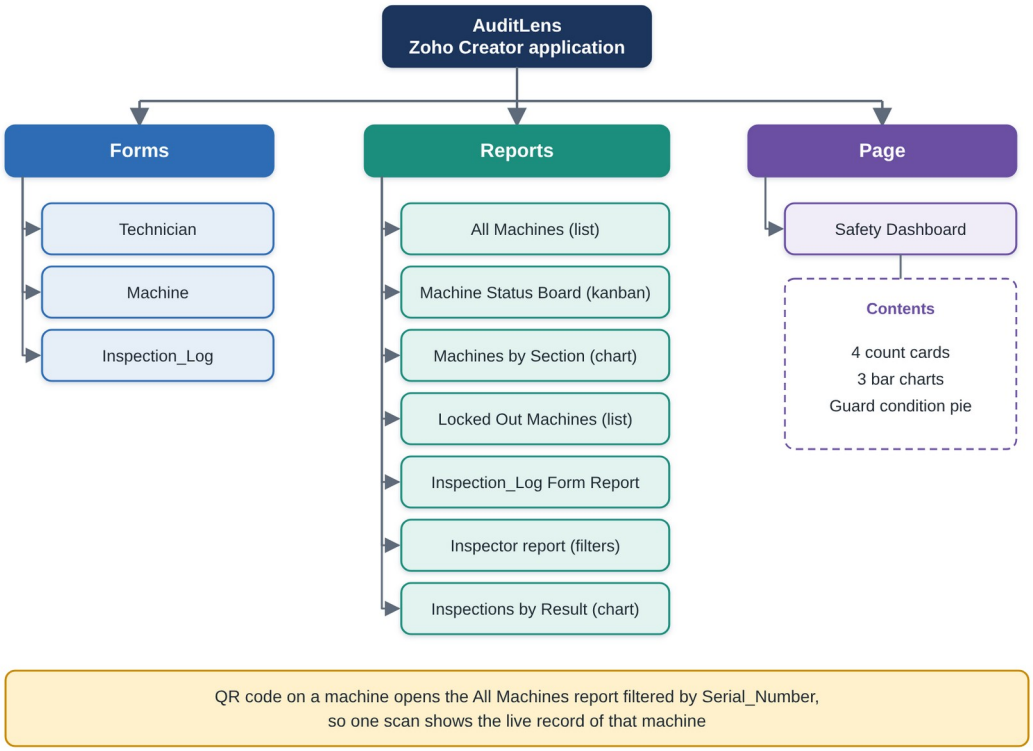


Figure 4.4: Navigation Structure of AuditLens

4.6 Access Control Design

In this version users sign in with their Zoho Creator accounts and the application is shared with named users. Table 4.3 proposes a role–permission matrix that can be configured as permission profiles in a future version (see Section 5.8).

Table 4.3: Proposed Role–Permission Matrix

Action	Administrator	Technician	Safety Manager	Auditor
Register / edit machines	Yes	No	View	View
Submit inspections	Yes	Yes	No	No
View dashboard	Yes	Yes	Yes	Yes
View inspection history	Yes	Own	Yes	Yes
Manage technicians	Yes	No	No	No

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in a Zoho Creator trial account as AuditLens (link name audit-lens). It uses Zoho Creator as the platform and Deluge for scripting. The menu contains the Technician, Machine and Inspection_Log forms with their reports, and the dashboard page named AUDIT LENS – DASHBOARD (Figure 5.1).

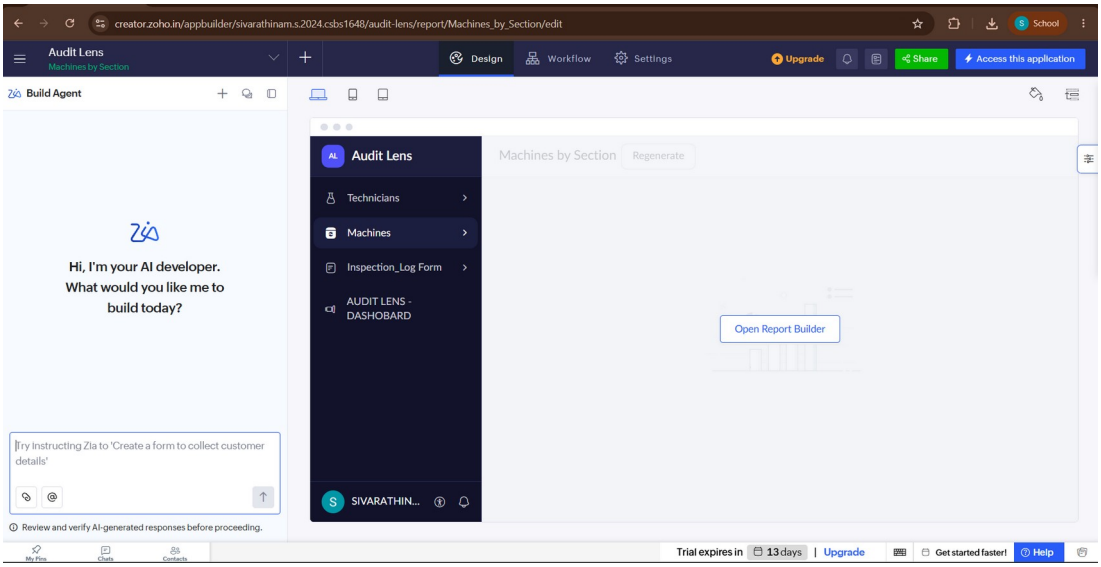


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Three forms capture all data: Technician, Machine and Inspection_Log (Table 4.2). Serial_Number is unique, so the same machine cannot be registered twice. The guard check on Inspection_Log is a Decision box, because a checkbox field showed the default values Choice 1, 2 and 3 (Table 6.2, BUG-05). Figures 5.2 and 5.3 show the Technician and Machine forms in the form builder, Figure 5.4 shows the Inspection_Log form in live mode and Figure 5.5 shows the Machine form in live mode. On the Machine form, Frequency_Days is labelled Inspection frequency and Last_Calibration_Date is labelled Last Inspection Date.

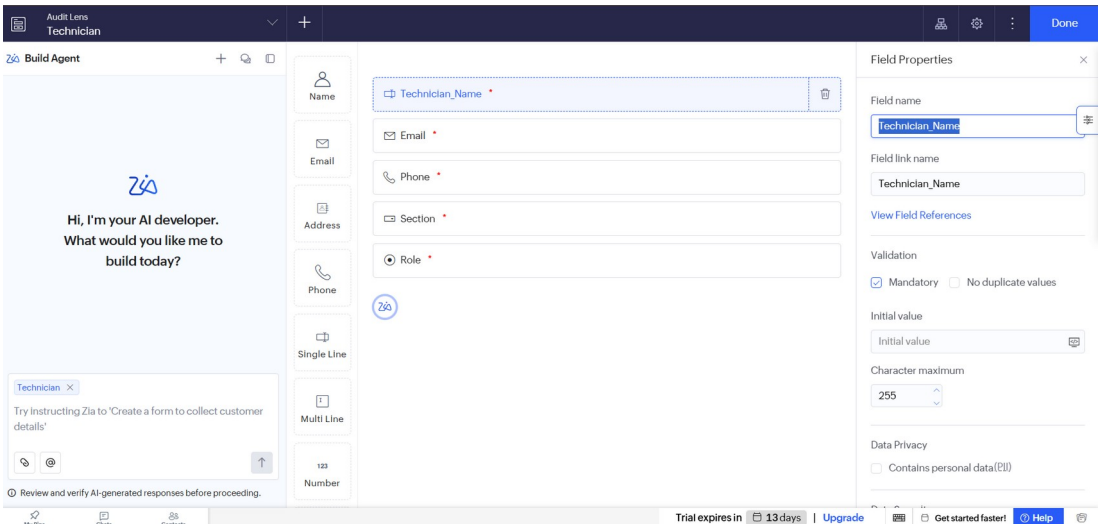


Figure 5.2: Technician Form in the Form Builder

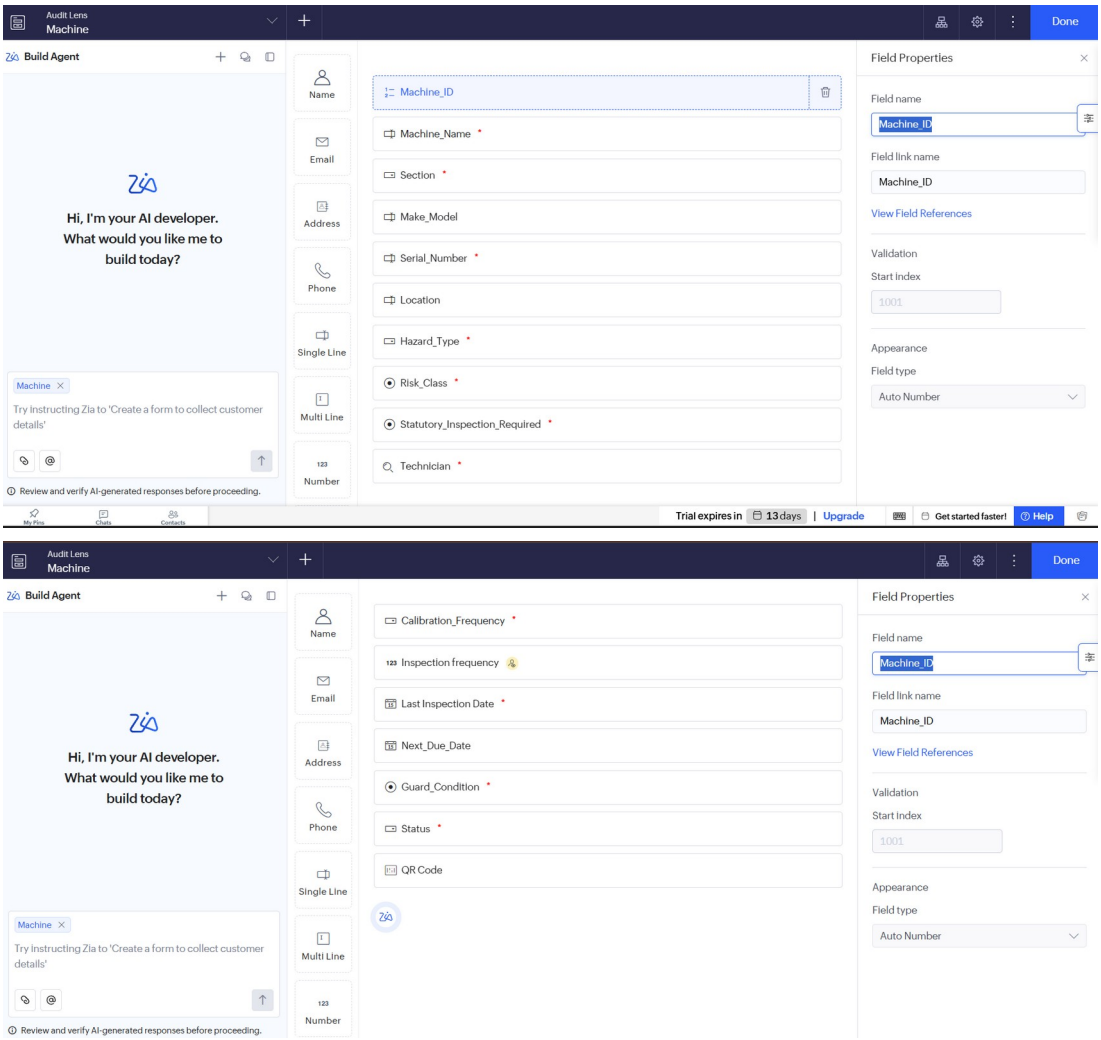


Figure 5.3: Machine Form in the Form Builder

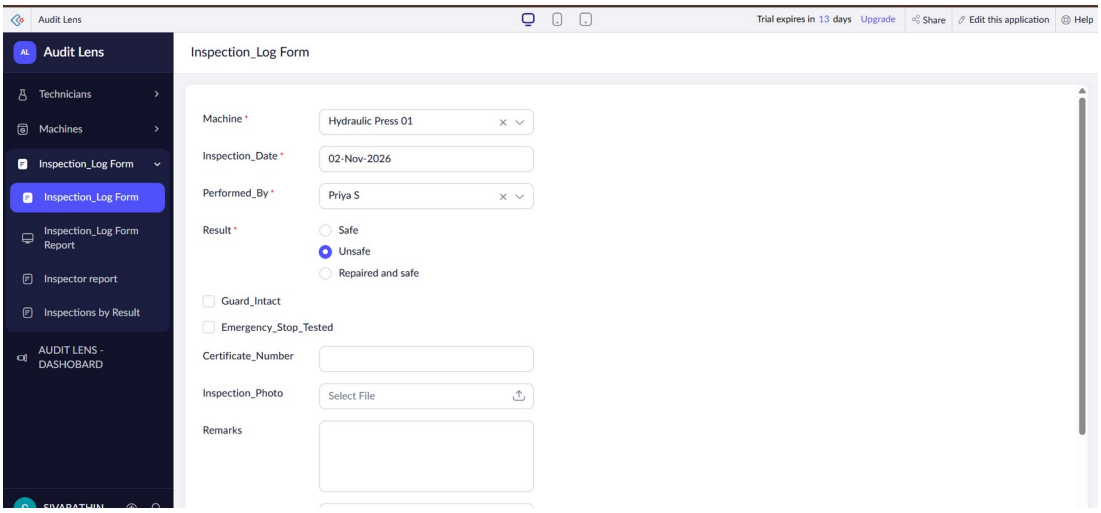


Figure 5.4: Inspection_Log Form in Live Mode

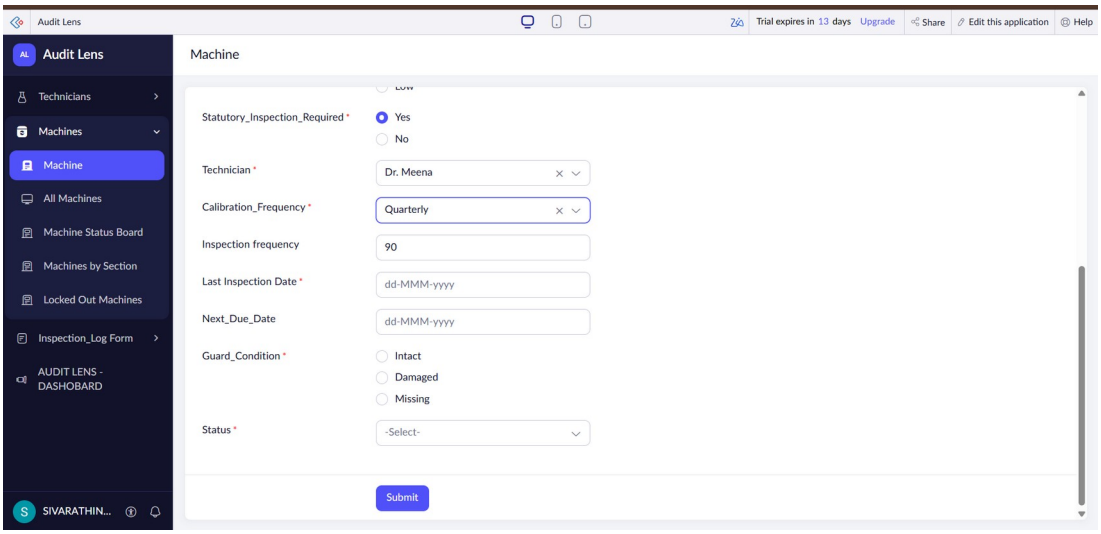


Figure 5.5: Machine Form in Live Mode

5.3 Reports

Table 5.1: Reports Implemented

Report	Type	Content
All Machines	List	Machine_ID, name, section, risk class, status, next due date, guard condition, QR code and technician; filters on Section, Status and Risk_Class
Machine Status Board	Kanban	Machines grouped by Status
Machines by Section	Chart	Count of machines in each section
Locked Out Machines	List	Machines whose Status is Locked Out
Inspection_Log Form Report	List	All inspections with a detail view of each record
Inspector report	List	Inspections with filters on Inspection_Date, Machine and Result, for audit review
Inspections by Result	Chart	Count of inspections by result

The All Machines report is also the target of the QR link: it is filtered by Serial_Number so a scan shows one machine (Section 5.12). The Inspector report gives an inspector the inspection history with filters (Figure 7.2).

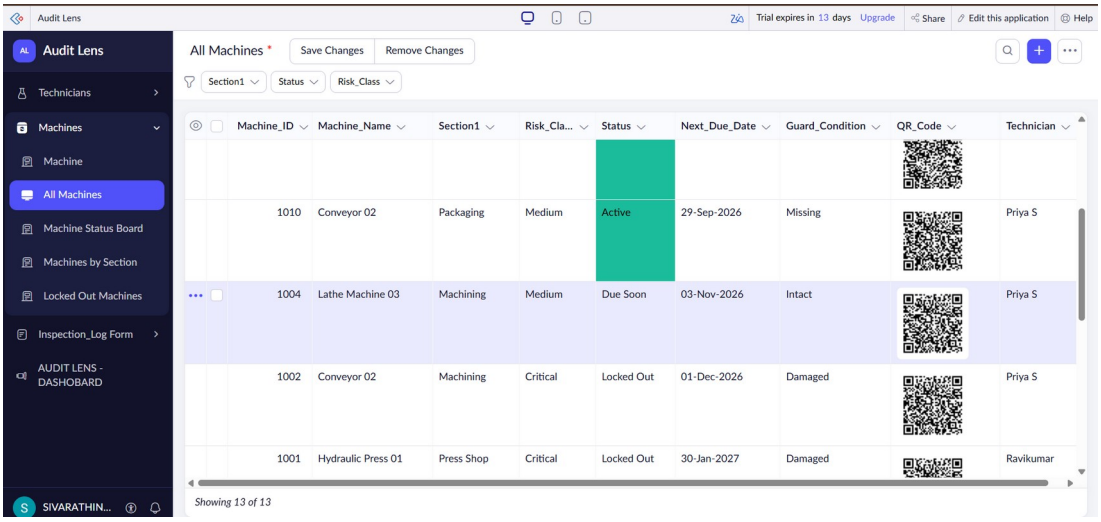


Figure 5.6: All Machines Report with QR Codes

5.4 Pages and Dashboards

The Safety Dashboard is a page built with the page builder. It has four count cards (Locked Out Machines, Due Soon Machines, Active Machines, Total Machines), charts of machines by Section, by Risk class and by Hazard type, and a pie chart of Guard condition. The chart reports are created as Pivot Charts and placed on the page as chart elements. On the test data the cards showed 2 locked out, 1 due soon, 10 active and 13 in total.

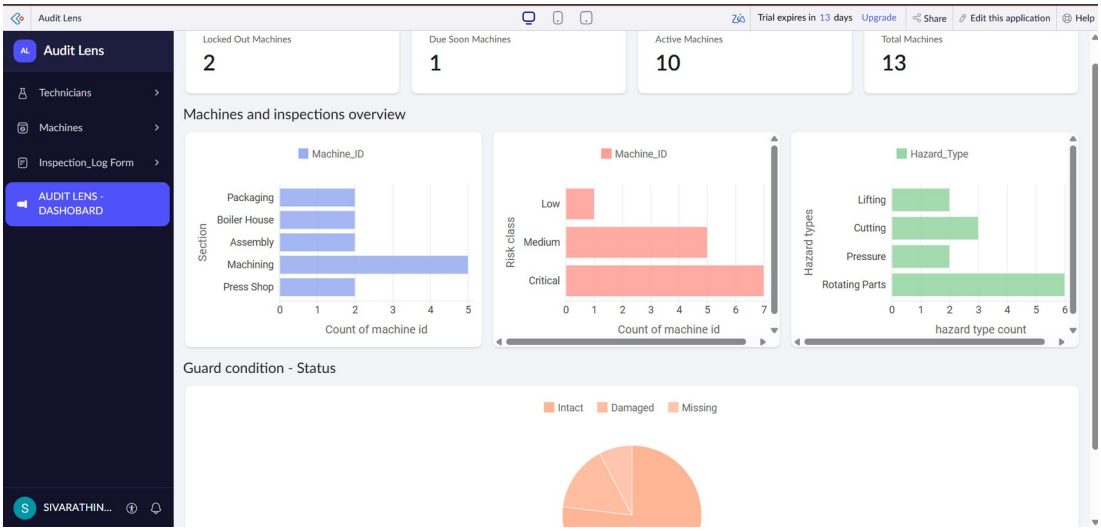


Figure 5.7: Safety Dashboard Page

5.5 Deluge Scripting

Deluge is used in the form workflows and in one schedule. Table 5.2 summarises the scripts. The key script (DS-04) is shown below and all scripts are listed in Appendix C.

Table 5.2: Deluge Scripts Implemented

ID	Script	Location	Purpose
DS-01	Set Frequency Days	Machine, On User Input of Calibration_Frequency	Monthly = 30, Quarterly = 90, Half-Yearly = 180, Yearly = 365
DS-02	Calculate Due Date and Lockout	Machine, On Validate	Sets Next_Due_Date; locks out if guard is Missing or Damaged
DS-03	Inspection Due Date	Inspection_Log, On Validate	Next_Due_Date = Inspection_Date + Frequency_Days
DS-04	Update Machine After Inspection	Inspection_Log, On Success	Updates machine dates, status and guard; sends e-mail on lockout
DS-05	Daily Safety Check	Schedule	Locks out overdue machines, marks machines due within 7 days as Due Soon and e-mails the technician

```
mach = Machine[ID == input.Machine];
mach.Last_Calibration_Date = input.Inspection_Date;
mach.Next_Due_Date = input.Inspection_Date.addDay(mach.Frequency_Days);
if(input.Result == "Unsafe" || input.Guard_Intact == false)
{
    mach.Status = "Locked Out";
    mach.Guard_Condition = "Damaged";
    tech = Technician[ID == mach.Technician];
    sendmail
    [
        from: zoho.adminuserid
        to: tech.Email
        subject: "Machine Locked Out: " + mach.Machine_Name
        message: "Machine " + mach.Machine_Name + " failed its inspection
and is locked out."
    ]
}
else
{
    mach.Status = "Active";
    mach.Guard_Condition = "Intact";
}
```

The Status value in the script must match the spelling in the Status drop-down exactly (for example "Locked Out"), otherwise the value is rejected.

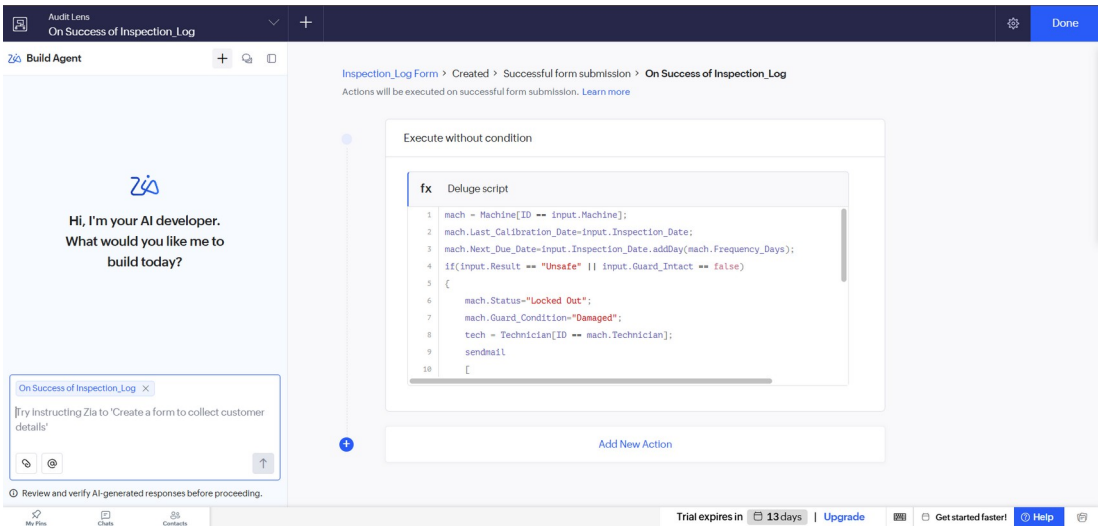


Figure 5.8: Deluge Script of Workflow 4 (Update Machine After Inspection)

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

#	Workflow	Trigger	What it Does	Status
1	Set Frequency Days	Machine, On User Input of Calibration_Frequency	Converts frequency to days	Tested
2	Calculate Due Date and Lockout	Machine, On Validate	Next_Due_Date = Last_Calibration_Date + Frequency_Days; lockout if guard Missing or Damaged	Tested
3	Inspection Due Date	Inspection_Log, On Validate	Next_Due_Date = Inspection_Date + machine's Frequency_Days	Tested
4	Update Machine After Inspection	Inspection_Log, On Success	Updates the machine; lockout and e-mail if unsafe	Tested
5	Daily Safety Check	Schedule (daily)	Locks out overdue machines; marks machines due within 7 days as Due Soon; e-mails the technician	Tested

Workflow 2 runs on validate because the record's own fields can be assigned with input. only in that event; the first attempt on the On Success event failed (Table 6.2, BUG-04). Workflow 5 is the Daily Safety Check schedule. When it runs, it locks out machines whose Next_Due_Date has passed, marks machines due within 7 days as Due Soon and e-mails the responsible technician (Figures 5.10 and 5.11).

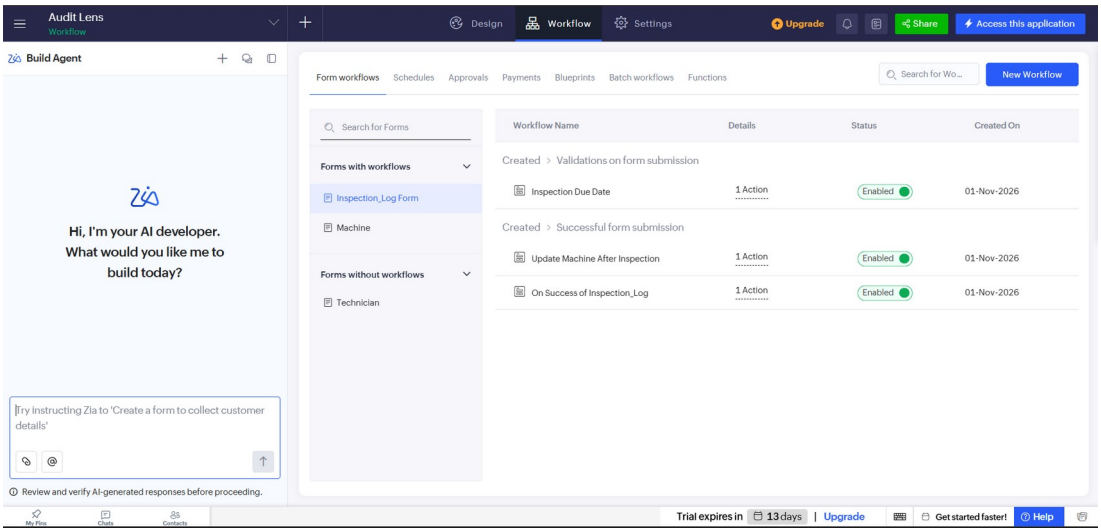


Figure 5.9: Form Workflows Configured on the Inspection_Log Form

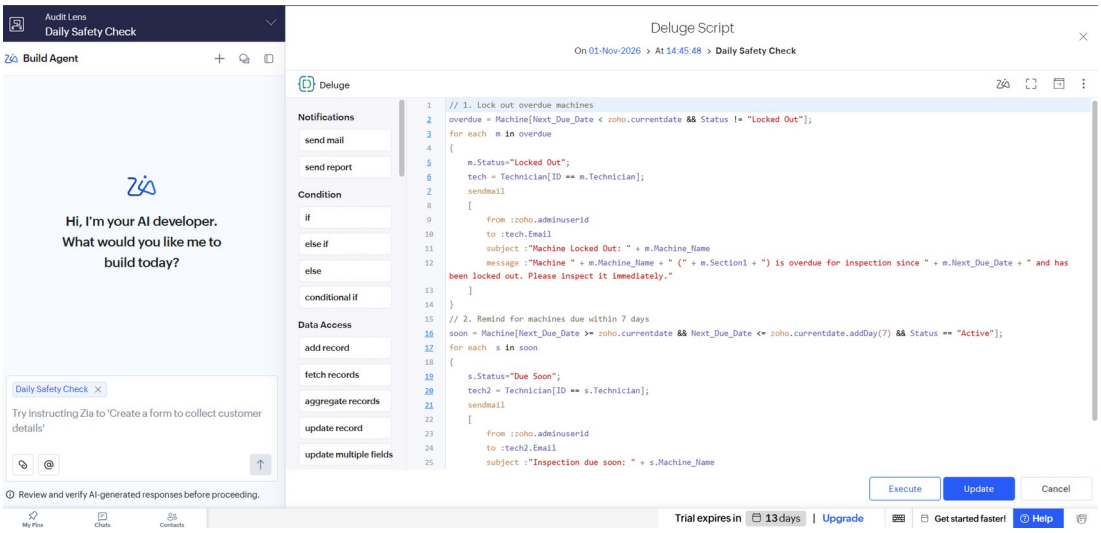


Figure 5.10: Daily Safety Check Schedule Script

5.7 Integrations

Table 5.4: Integrations Implemented

Service	Use	Mechanism
Zoho Mail	Lockout alert e-mail to the assigned technician	Deluge sendmail task in Workflow 4
External REST API	Not needed	—

No external API integration was required. The e-mail is sent from a Zoho trial sender address (Section 7.3).

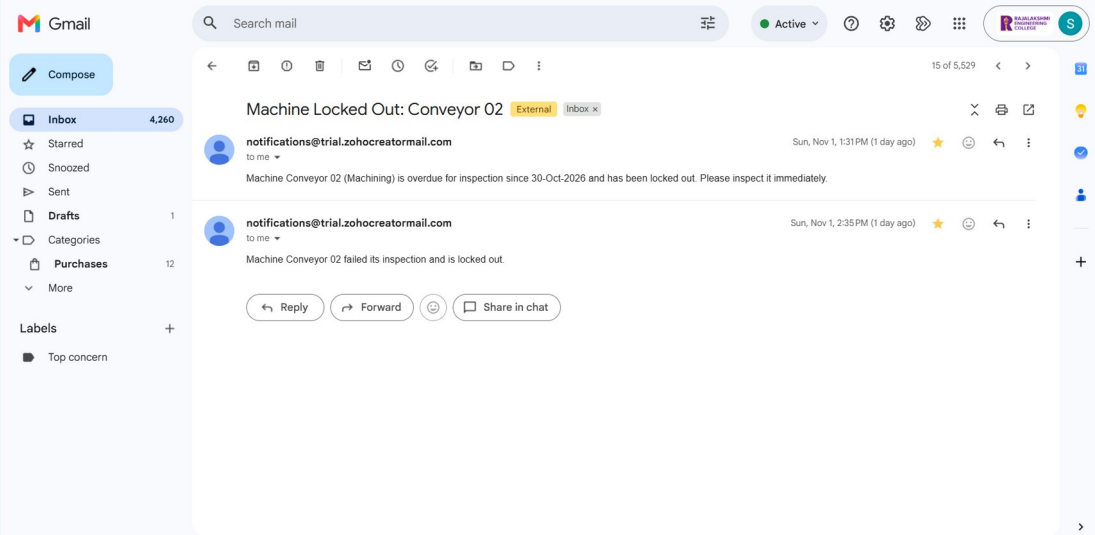


Figure 5.11: Lockout E-mail Received by the Technician

5.8 User Management and Role-Based Access

Users sign in with their Zoho accounts, and the administrator shares the application with named users. Technicians are also stored as records in the Technician form, so every machine and inspection is linked to a responsible person. Permission profiles for Technician, Safety Manager and Auditor, and role-based dashboards, are proposed as an extension (Section 4.6 and Chapter 9).

5.9 AI and Analytics Features

No AI features (Zia or the AI application builder) were used. Analytics is provided by the dashboard cards and charts described in Section 5.4. Zoho Analytics integration is a future enhancement.

5.10 Pro-Code Extension

No pro-code extension (widgets, Catalyst or custom functions) was required. All logic fits in Deluge form workflows.

5.11 Mobile and Portal Access

The QR code is scanned with a phone camera and opens the application link in the phone browser. The scanner must log in to the application. A public Safe to Operate status page is listed as future work.

5.12 QR Code Design

The QR code encodes a link to the All Machines report filtered by Serial_Number:

```
https://creatorapp.zoho.in/sivarathinam.s.2024.csbs1648/audit-lens/
#Report:All_Machines?Serial_Number=${Serial_Number}
```

- The QR stores a link, not a status, so a scan always shows current data and the label never needs reprinting.
- Serial_Number is used instead of the auto-number Machine_ID because the auto-number does not exist until the record is saved (Table 6.2, BUG-03).

- Limitation: the person scanning needs a login to the application.

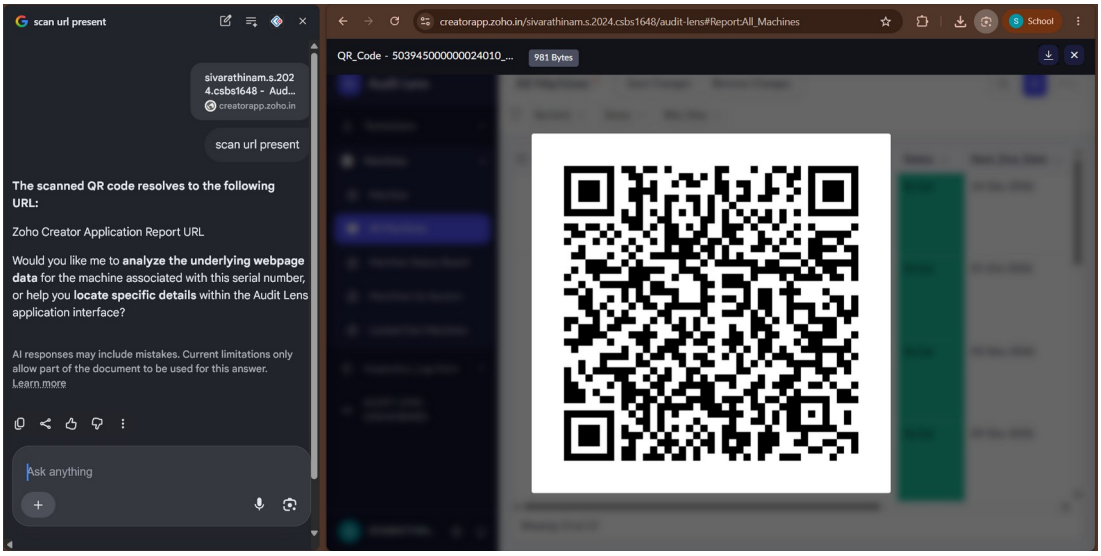


Figure 5.12: QR Code Scanned and Resolved to the Application URL

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was done at three levels. Unit testing checked each Deluge workflow on its own by saving a machine or an inspection and reading the resulting fields. Integration testing checked the chain from an inspection submission to the machine update and the e-mail. Report testing checked that the QR scan and the dashboard cards show the real data. Test data of 13 machines was used.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Expected Result	Actual Result	Status
TC-01	FR-03, 04	Add a Quarterly machine	Frequency_Days = 90; Next_Due_Date calculated	As expected	Pass
TC-02	FR-05	Save a machine with guard Missing	Status becomes Locked Out	As expected	Pass
TC-03	FR-07, 08	Submit an Unsafe inspection	Machine Locked Out, guard Damaged, dates updated	As expected	Pass
TC-04	FR-09	Check the lockout e-mail	E-mail received	As expected	Pass
TC-05	FR-10	Submit a Safe or Repaired inspection with guard ticked	Machine returns to Active	As expected	Pass
TC-06	FR-11	Scan the QR code	Opens the machine record with live data	As expected	Pass
TC-07	FR-12	Open the dashboard cards	Show real counts	13 machines counted	Pass
TC-08	FR-01	Bulk import of 10 machines	All rows import	5 imported, 5 rejected (value spelling mismatch); all rows imported after the fix	Pass (after fix)
TC-09	FR-14	Run the Daily Safety Check schedule	Overdue machine locked out and e-mailed; machine due within 7 days set to Due Soon	Conveyor 02 locked out and e-mailed; Lathe Machine 03 shown as Due Soon	Pass

6.3 Debugging Deluge Scripts

Errors were located from the messages shown by the Deluge editor and at run time, and fixed by matching the script to the form. Most errors came from field link names that

differ from field labels. Figure 6.1 shows the error "Variable Section does not exist in Machine", which was fixed by using the field link name Section1. The defects found this way are listed in Table 6.2.



Figure 6.1: Deluge Error Message Seen During Debugging

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	"Variable not defined" error in a script	Field link name differed from the field label	Checked and matched the Field link name	Closed
BUG-02	"Variable Section does not exist"	Two forms had a Section field, so Creator named the second one Section1	Used Section1 in the script	Closed
BUG-03	QR input error	Auto-number ID is not available before saving	Used Serial_Number, inserted with Insert fields	Closed
BUG-04	Record's own fields could not be edited on success	Wrong event for input. assignments	Used On Validate for the date calculation	Closed
BUG-05	Checkbox showed Choice 1, 2, 3	Wrong field type	Replaced with a Decision box	Closed
BUG-06	Import rejected some rows	Status and frequency values did not match the drop-down spelling ("Locked out" in the file vs "Locked Out" in the drop-down, "Half-Yearly")	Matched the spelling, then re-imported	Closed
BUG-07	Dashboard cards showed Ct ()	Designer shows a placeholder; criteria were incomplete	Set function, form and criteria; checked in Access mode	Closed

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-08	Chart report type missing	Creator names it Pivot Chart	Used Pivot Chart and page chart elements	Closed

6.5 User Acceptance Testing

The application was checked against acceptance criteria drawn from the objectives, using the test data of 13 machines and the technician records. Table 6.3 lists each criterion, the evidence and the outcome.

Table 6.3: User Acceptance Criteria and Outcome

Acceptance Criterion	Evidence	Outcome
A technician can open a machine by scanning its QR code	TC-06; Figure 5.12	Met
Due dates are calculated without manual entry	TC-01; Figure 5.5	Met
An unsafe or guard-less machine is locked out automatically	TC-02, TC-03; Figure 5.6	Met
The responsible technician receives an e-mail alert	TC-04; Figures 5.11, 8.2	Met
Overdue machines are locked out and near-due machines are marked without manual action	TC-09; Figure 5.10	Met
An inspector can review the inspection history	Figure 7.2	Met
The dashboard shows live counts	TC-07; Figure 5.7	Met

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

AuditLens was developed and tested in the Development environment of a Zoho Creator trial account. It was then published to the Stage environment and to the Production environment as version 1.0 on 02 November 2026 (Figure 7.1). Table 7.1 shows the deployment checklist.

Table 7.1: Deployment Checklist

Item	Status
Forms, reports and dashboard built	Done
Form workflows tested	Done
Daily Safety Check schedule tested	Done
QR codes generated for machines	Done
Published to Stage (version 1.0)	Done
Published to Production (version 1.0)	Done
Role-based permission profiles	Proposed for a future version

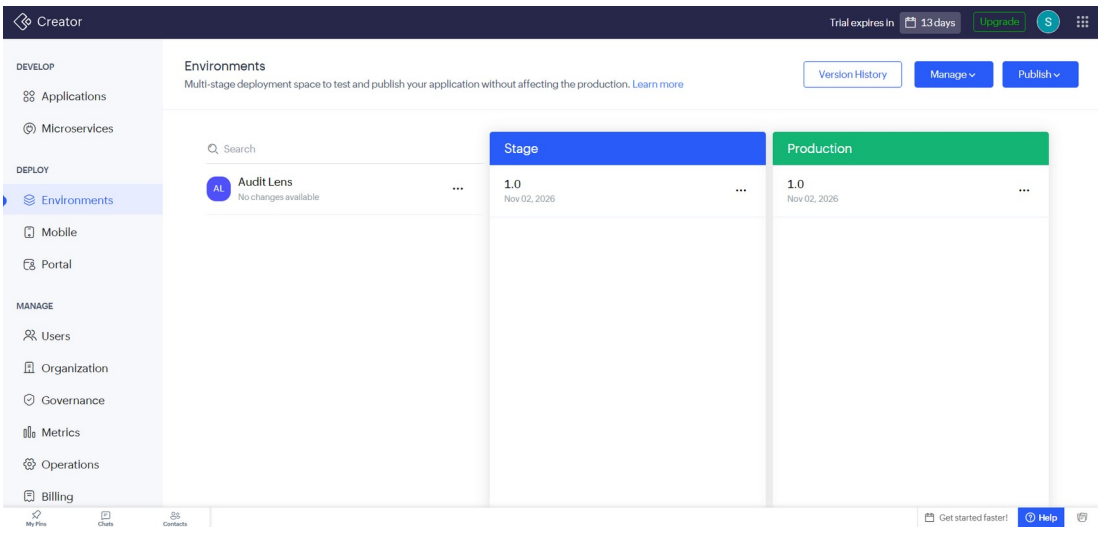


Figure 7.1: Environments Showing Version 1.0 in Stage and Production

7.2 Performance Optimisation

The following design choices keep the application light.

Table 7.2: Performance Design Considerations

Design choice	Benefit
Dates and status calculated by workflows once, at save time	Reports read stored values and do not recalculate

Design choice	Benefit
QR link filtered by Serial_Number	A scan loads one record, not the whole list
Three small forms with lookups	No duplicated machine data in the inspection log
Dashboard counts from criteria on the Machine form	Live counts without custom scripts

7.3 Security Features

Table 7.3: Security Controls Implemented

Control	How it is applied
Login required	Scanning the QR code requires a login to the application
Unique Serial_Number	Prevents duplicate machine records
Mandatory fields and fixed drop-downs	Prevent incomplete or misspelled data
Workflow-controlled status	Status and dates are set by workflows, not typed
Controlled access	The application is shared only with named users who sign in with Zoho accounts

Limitations: the alert e-mail is sent from a Zoho trial sender address, and trial account limits apply.

7.4 Governance and Administration

Each inspection is submitted as a new Inspection_Log record with date, technician, result and photo, so the history serves as audit evidence. The Inspector report lets an inspector filter the history by inspection date, machine and result (Figure 7.2). Application, form and field names follow one pattern (for example Machine, Inspection_Log, Serial_Number).

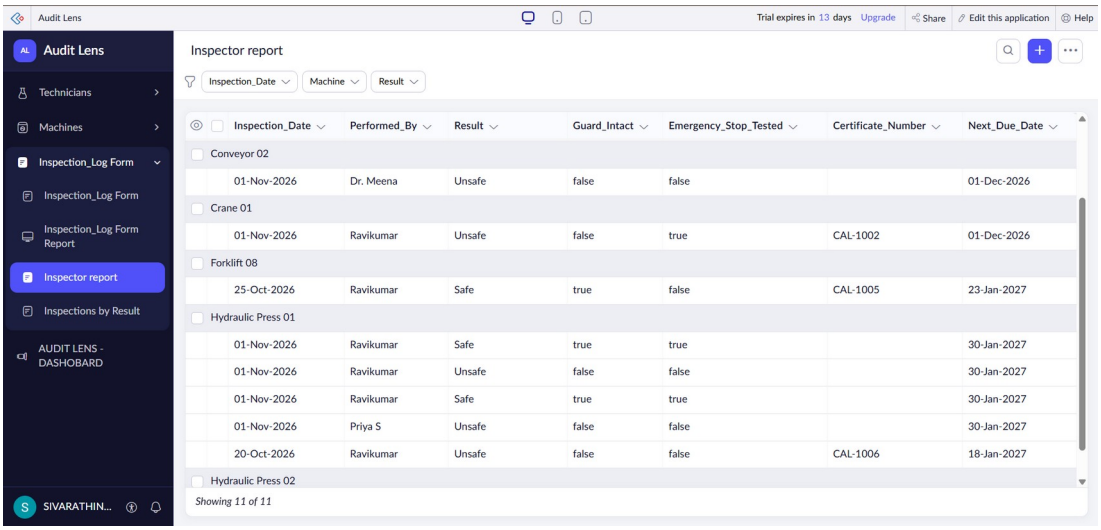


Figure 7.2: Inspector Report Showing Inspection History for Audit

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

Scenario: a technician scans the QR code on Hydraulic Press 01 and opens the Inspection_Log form. The technician selects the machine, enters the inspection date, chooses the result Unsafe and leaves Guard_Intact unticked (Figure 5.4). On submission the inspection is stored with its details (Figure 8.1). The machine's dates are updated, its Status becomes Locked Out and its Guard_Condition becomes Damaged (Figure 5.6). The assigned technician receives the lockout e-mail (Figure 8.2), and the dashboard counts show the locked out machines (Figure 8.3). A later safe inspection with the guard ticked returns the machine to Active.

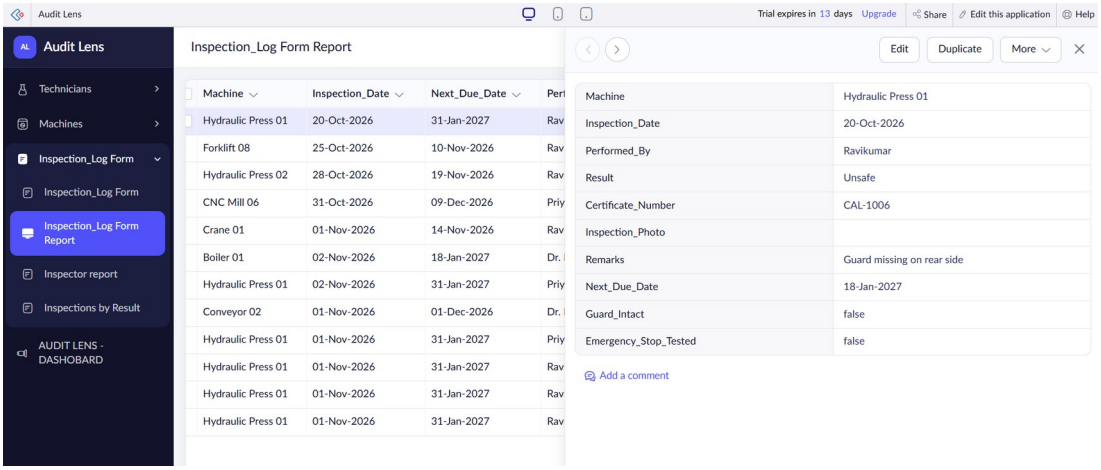


Figure 8.1: Stored Inspection Record in the Inspection_Log Form Report

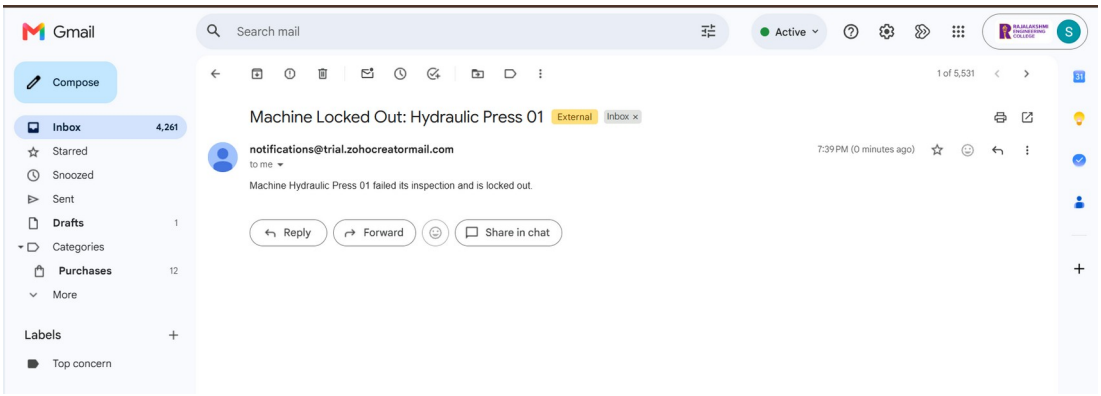


Figure 8.2: Lockout E-mail for Hydraulic Press 01

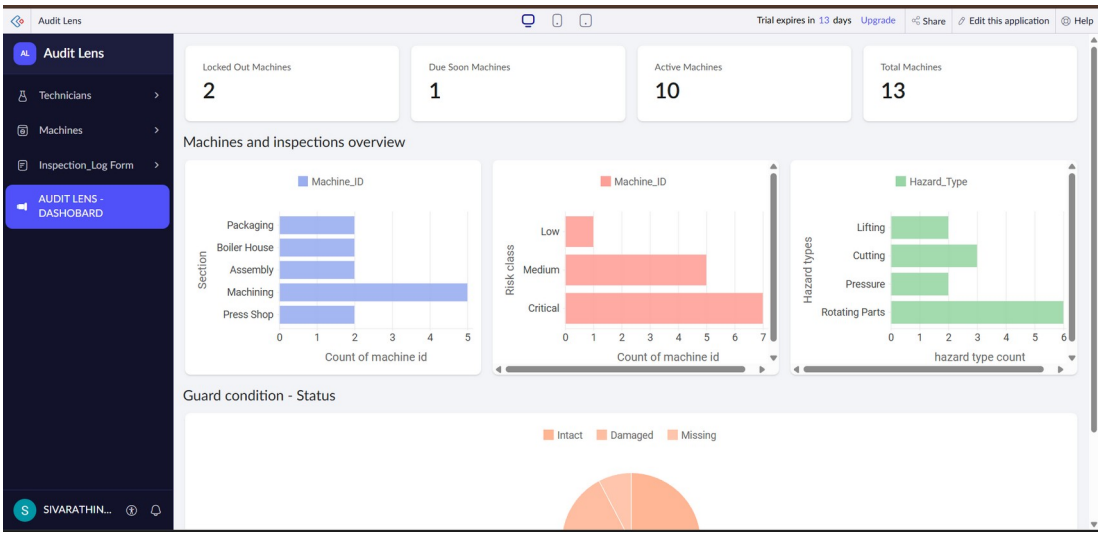


Figure 8.3: Dashboard After the Lockout

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective	Outcome	Status
Register every machine once with a unique QR tag	Machine form with unique Serial_Number and QR_Code field	Achieved
Calculate inspection due dates automatically	Workflows 1 to 3 calculate Frequency_Days and Next_Due_Date	Achieved
Lock out unsafe, guard-less or overdue machines	Unsafe and guard-less lockout (form workflows) and overdue lockout (Daily Safety Check schedule) tested	Achieved
Notify the technician by e-mail	Lockout e-mail received in testing	Achieved
Store inspection history as audit evidence	Inspection_Log keeps every inspection; the Inspector report filters the history	Achieved
Show safety status on a dashboard	Four cards and four charts show live data	Achieved

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Aspect	Paper and Excel	AuditLens
Due dates	Written by hand	Calculated from frequency
Reminders	None	Daily Safety Check schedule marks Due Soon and e-mails the technician
Missing guard	Noticed by chance	Machine locked out automatically
Alerts	None	E-mail to the assigned technician
Machine identity	Name in a sheet	QR code linking to the live record

Aspect	Paper and Excel	AuditLens
Audit proof	Records searched by hand	Inspection history with date, result and photo
Overview	Counted by hand	Live dashboard

8.4 Limitations of the Low-Code Approach

- QR scanning requires a login to the application.
- Due Soon and overdue statuses are updated when the daily schedule runs, not at the moment a date passes.
- Trial account limits apply.
- The e-mail sender is a Zoho trial address.
- Permission profiles by role and a public status page are not part of this version.
- The lockout is a software status; it does not stop the physical machine.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Forms, reports, charts and the QR field were built visually (no-code). The due-date and lockout rules needed Deluge scripts (low-code). Bugs such as field link names and event timing showed that even a small application needs an understanding of how the platform works. Features that the platform does not provide directly, such as a physical lockout through IoT sensors, would need pro-code extensions.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

AuditLens replaces paper and Excel tracking with a QR-based system that calculates due dates, locks out unsafe machines automatically, alerts the responsible person and keeps inspection evidence ready for an inspector. Three forms, form workflows, a daily scheduled workflow, a set of reports and a dashboard were built and tested on Zoho Creator and published as version 1.0 to Stage and Production. The project shows how a low-code platform can address a real safety and compliance problem.

9.2 Future Enhancements

- WhatsApp and SMS alerts.
- Role-based dashboards for Technician, Safety Manager and Auditor.
- Zoho Analytics integration.
- Predictive maintenance from failure history.
- IoT sensor data.
- Mobile offline mode.
- A public “Safe to Operate” status page.
- Scheduled e-mail summary of overdue and due-soon machines for the Safety Manager.

9.3 Learning Outcomes and Career Relevance

This project developed skills in data modelling, form and report design, page building, Deluge scripting, workflow automation, QR code design and systematic debugging. These skills apply to roles such as low-code developer, Zoho Creator consultant, business analyst and citizen developer.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created and menu organised	Y	AuditLens; forms, report, dashboard	5.1	Fig. 5.1
II	Forms with 8 or more distinct field types	Y	Machine, Inspection_Log	5.2	Table 4.2
II	Lookup field (relationship between forms)	Y	Technician, Machine, Performed_By	4.2, 5.2	Table 4.1
II	Subform (parent–child data)	N	Not needed	4.2	—
II	Field validations / unique field	Y	Unique Serial_Number; mandatory fields	5.2	Table 4.2
II	List report	Y	All Machines, Inspector report	5.3	Table 5.1
II	Kanban / Calendar / Timeline / Map report	Y	Machine Status Board (kanban)	5.3	Table 5.1
II	Pivot table / chart report	Y	Pivot Charts on dashboard	5.4	Fig. 5.7
II	Page / dashboard built with page builder	Y	Safety Dashboard	5.4	Fig. 5.7
II	Deluge – form events (validate / success / user input)	Y	On User Input, On Validate, On Success	5.5	Table 5.2
II	Deluge – conditions, fetch, field updates	Y	Machine fetch and update	5.5	App. C
II	E-mail notification	Y	Lockout e-mail (sendmail)	5.7	Fig. 5.11
II	Form workflow	Y	4 form workflows	5.6	Table 5.3
II	Scheduled workflow	Y	Daily Safety Check	5.6	Fig. 5.10
II	Approval process	N	Not required	5.6	—
II	Blueprint	N	Not required	5.6	—
II	Batch / report workflow	N	Not required	5.6	—
II	Testing and debugging evidence	Y	9 test cases, 8 defects	6	Tables 6.1, 6.2
III	Integration with a Zoho service	Y	Zoho Mail	5.7	Table 5.4
III	Connection / REST API call (invokeurl)	N	Not needed	5.7	—
III	Users, roles and permission profiles	N	Proposed (Table 4.3)	5.8	—
III	Zia / AI feature	N	Not used	5.9	—
III	Analytics (charts / cards)	Y	Dashboard cards and charts	5.9	Fig. 5.7

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
IV	Deployment across environments	Y	Development, Stage, Production (v1.0)	7.1	Fig. 7.1
IV	Security controls	Y	Login, unique serial, mandatory fields	7.3	Table 7.3
IV	Governance (audit trail, naming)	Y	Inspection history; Inspector report	7.4	Fig. 7.2
V	Pro-code extension (widgets / Catalyst)	N	Not required	5.10	—
V	Low-code limitations and spectrum analysis	Y	Sections 8.4, 8.5	8.4, 8.5	—
—	Mobile app / portal access	N	Not used; QR opens the link in a phone browser	5.11	Fig. 5.12

Total features implemented: 20 / 29

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3

APPENDIX C – COMPLETE DELUGE SOURCE CODE

Table 5.2 lists the five Deluge scripts used in AuditLens. Each script is shown below with its location.

C.1 Machine form – DS-01 and DS-02

```
// Script ID : DS-01   Workflow: Set Frequency Days
// Location  : Machine -> on user input of Calibration_Frequency
if(input.Calibration_Frequency == "Monthly")
{
    input.Frequency_Days = 30;
}
else if(input.Calibration_Frequency == "Quarterly")
{
    input.Frequency_Days = 90;
}
else if(input.Calibration_Frequency == "Half-Yearly")
{
    input.Frequency_Days = 180;
}
else if(input.Calibration_Frequency == "Yearly")
{
    input.Frequency_Days = 365;
}

// Script ID : DS-02   Workflow: Calculate Due Date and Lockout
// Location  : Machine -> on validate
if(input.Calibration_Frequency == "Monthly")
{
    input.Frequency_Days = 30;
}
else if(input.Calibration_Frequency == "Quarterly")
{
    input.Frequency_Days = 90;
}
else if(input.Calibration_Frequency == "Half-Yearly")
{
    input.Frequency_Days = 180;
}
else if(input.Calibration_Frequency == "Yearly")
{
    input.Frequency_Days = 365;
}
input.Next_Due_Date =
input.Last_Calibration_Date.addDay(input.Frequency_Days);
if(input.Guard_Condition == "Missing" || input.Guard_Condition ==
"Damaged")
{
    input.Status = "Locked Out";
}
```

C.2 Inspection_Log form – DS-03 and DS-04

```
// Script ID : DS-03   Workflow: Inspection Due Date
// Location  : Inspection_Log -> on validate
mach = Machine[ID == input.Machine];
input.Next_Due_Date = input.Inspection_Date.addDay(mach.Frequency_Days);

// Script ID : DS-04   Workflow: Update Machine After Inspection
// Location  : Inspection_Log -> on success
mach = Machine[ID == input.Machine];
mach.Last_Calibration_Date = input.Inspection_Date;
mach.Next_Due_Date = input.Inspection_Date.addDay(mach.Frequency_Days);
if(input.Result == "Unsafe" || input.Guard_Intact == false)
{
    mach.Status = "Locked Out";
    mach.Guard_Condition = "Damaged";
    tech = Technician[ID == mach.Technician];
    sendmail
    [
        from: zoho.adminuserid
        to: tech.Email
        subject: "Machine Locked Out: " + mach.Machine_Name
        message: "Machine " + mach.Machine_Name + " failed its inspection
and is locked out."
    ]
}
else
{
    mach.Status = "Active";
    mach.Guard_Condition = "Intact";
}
```

C.3 Daily Safety Check schedule – DS-05

```
// Script ID : DS-05   Workflow: Daily Safety Check
// Location  : Schedule (daily)
// 1. Lock out overdue machines
overdue = Machine[Next_Due_Date < zoho.currentdate && Status != "Locked
Out"];
for each m in overdue
{
    m.Status = "Locked Out";
    tech = Technician[ID == m.Technician];
    sendmail
    [
        from: zoho.adminuserid
        to: tech.Email
        subject: "Machine Locked Out: " + m.Machine_Name
        message: "Machine " + m.Machine_Name + " (" + m.Section1 + ") is
overdue for inspection since " + m.Next_Due_Date + " and has been locked
out. Please inspect it immediately."
    ]
}
// 2. Remind for machines due within 7 days
soon = Machine[Next_Due_Date >= zoho.currentdate && Next_Due_Date <=
zoho.currentdate.addDay(7) && Status == "Active"];
for each s in soon
{
```

```
    s.Status = "Due Soon";
    tech2 = Technician[ID == s.Technician];
    sendmail
    [
        from: zoho.adminuserid
        to: tech2.Email
        subject: "Inspection due soon: " + s.Machine_Name
        message: "Machine " + s.Machine_Name + " is due for inspection on "
+ s.Next_Due_Date + ". Please plan the inspection."
    ]
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application link shared by the administrator (https://creatorapp.zoho.in/sivarathinam.s.2024.csbs1648/audit-lens/) and sign in with your Zoho account. The left menu shows the Technician, Machine and Inspection_Log forms with their reports (such as All Machines, Machine Status Board, Locked Out Machines and Inspector report) and the dashboard. To open one machine, scan the QR code on it with a phone camera and sign in.

D.2 Instructions for Technicians

- Scan the QR code on the machine. Its live record opens with status, due date and guard condition.
- If the status is Locked Out, do not operate the machine. Report to the safety manager.
- Open Inspection_Log and choose the machine, the inspection date and your name.
- Select the Result, tick Guard_Intact only if the guard is fitted and undamaged, tick Emergency_Stop_Tested, enter the certificate number, attach a photo and add remarks.
- Click Submit. The machine’s dates and status update automatically. If the result is Unsafe or the guard is not intact, an e-mail alert is sent.

D.3 Instructions for the Administrator

- Add technicians in the Technician form.
- Add each machine in the Machine form: enter a unique Serial_Number, choose the Calibration_Frequency and the responsible technician. Frequency_Days and Next_Due_Date fill automatically.
- Print the QR code from the All Machines report and attach it to the machine.
- Use the Safety Dashboard to see locked out, due soon, active and total machines.
- Open the Inspector report to review inspection history by date, machine and result.
- Use the spelling from the Status drop-down when importing data (for example "Locked Out", "Half-Yearly").